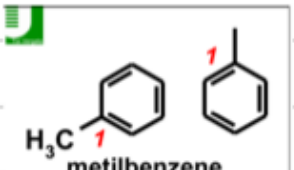
$-CH=CH_2$ VINILE (etenile)

$-CH_2CH=CH_2$ ALLILE (2-propenile)



metilbenzene

toluene

(nome corrente accettato)

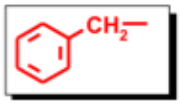
Il gruppo monovalente che si ottiene togliendo un H al benzene si chiama fenile

C_6H_5- = Ph = → **fenile**

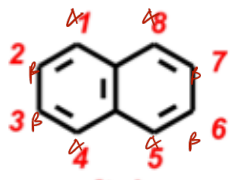
benzene con una valenza libera

Phenyl

ATTENZIONE!!! → non si confonda il fenile (gruppo con valenza libera su un C sp^2) con il benzile, gruppo con una valenza libera sul C sp^3, proveniente dal metilbenzene.

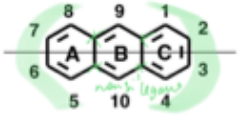


→ benzile benzene con gruppo CH_2



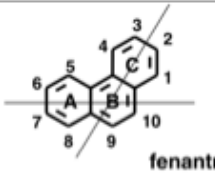
naftalene

Tre anelli benzenici condensati → $C_{14}H_{10}$ → 2 isomeri strutturali



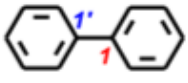
antracene

prima cicli esterni



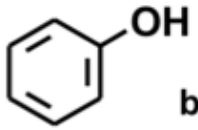
fenantrene

Due anelli benzenici collegati da un legame singolo (e quindi non hanno C in comune)



bifenile

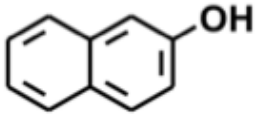
La numerazione viene fatta separatamente sui due anelli, usando per uno dei due anelli i numeri con apice, dando i numeri 1 e 1' ai C legati tra loro.



molto arcaico

benzenolo

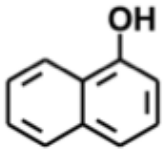
fenolo (*)



2-naftalenolo

2-naftolo (*)

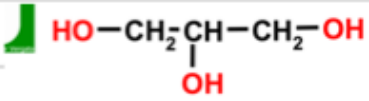
∘ β-naftolo

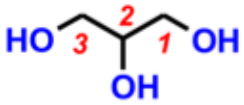


1-naftalenolo

1-naftolo (*) ∘ α-naftolo

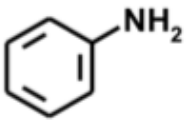
(*) nome corrente accettato





1,2,3-propantriolo

glicerolo (nome corrente accettato)

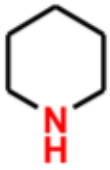


benzenammina

anilina(*)

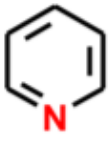
azaciclopentano

pirrolidina(*)



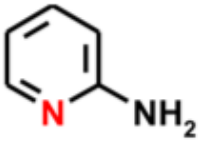
azacicloesano

piperidina(*)



azabenzene

piridina(*)



2-amminoazabenzene

2-amminopiridina

OH - C sp^3 ALCOOL -OLO IDROSSI- / ALCOOL -ICO (semplici)

COH ossidrile -C-OH ≠ CHO aldeidico C=O H

OH - C sp^2 aromatico FENOLO

OH - C sp^2 alchene ENOLO

SH - TIOLO -TIOLO SOLFANIL-

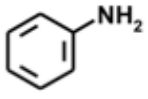
-O-R ETERE -ETERE alchilOSSI-

-O- ETERI CICLICI OSSA-

-S-R SOLFURI -SOLFURO alchilSOLFANIL-

-S- SOLFURI CICLICI TIA-

N -NH- $-NH_2$ AMMINE -AMMINA -AMMINO



benzenammina
anilina(*)

=N- N H AMMINE CICLICHE AZA-

SALI DI AMMONIO

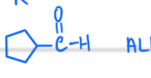
OSSONIO

SOLFONIO

DIAZONIO

SOLFOSSIDI
SOLFONI
ACIDI SOLFONICI

H, R C=O ALDEIDE -ALE / ALDEIDE -OICA



ALDEIDE C del gruppo aldeide NON FA PARTE del ciclo -CARBALDEIDE

R-C(=O)-R CHETONE -ONE -OSSO

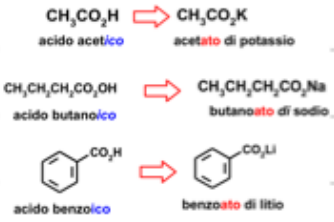
CH_3-C(=O)- GRUPPO ACETILE

GRUPPI ACETILICI

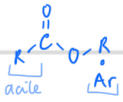
$-CO_2H$ ACIDO CARBOSSILICO acido -oico carbossi-

$Ar-CO_2H$ acido -carbossilico

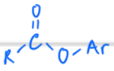
(*) acido benzoico
acido benzencarbossilico



$-CO_2M$ SALE DEGLI ACIDI CARBOSSILICI -oato di metallo metallo carbossilato



$-CO_2R$ ESTERE -oato di alchile alchiloossicarbonil
aril

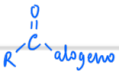


aril carbossilato

(*) benzoato di
~~benzencarbossilato di~~

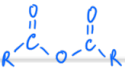
$C(O)NH_2$ AMMIDE -ammide carbammoil-

$Ar-CONH_2$ -carbossiammide



$-C(O)A$ ALOGENURO ACILICO alogenuro di alchiloile alogenocarbonil- o alogenoformil

$Ar-C(O)X$ alogenuro di arilcarbonile



$-C(O)O(O)C-$ ANIDRIDE anidride -oica 2 in ordine alfabetico

$Ar-C(O)O(O)C$ anidride -carbossilica

$C\equiv N$ $-CN$ NITRILE -nitrile ciano-

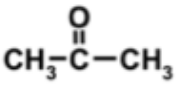
$Ar-CN$ -carbonitrile

ACILE alchiloil

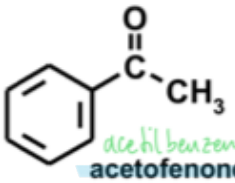
$Ar-\overset{O}{\overset{\|}{C}}-$ arilicocarbonil (*) benzoil
~~benzenecarbonil~~

- ESTERI SOLFONICI
- SOLFATI
- ALOGENURI DI SOLFONILE
- SOLFONAMMIDE

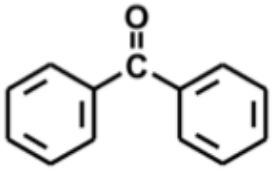
Alcuni nomi correnti sono ancora accettati:



acetone(*)
propanone
dimetil chetone



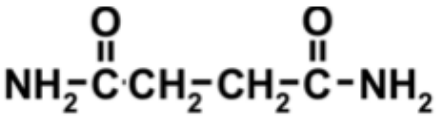
acetilbenzene anche
acetofenone(*)
feniletanone
fenil metil chetone



benzofenone(*)
difenil chetone

Acidi Monocarbossilici	Nome sistematico	Nome corrente
$H\text{-}CO_2H$	acido metanoico	acido formico
CH_3CO_2H	acido etanoico	acido acetico
$CH_3CH_2CO_2H$	acido propanoico	acido propionico
$CH_3CH_2CH_2CO_2H$	acido butanoico	acido butirrico
$CH_3(CH_2)_3CO_2H$	acido pentanoico	acido valerico
$(CH_3)_3CCO_2H$	acido 2,2-dimetilpropanoico	acido pivalico
$CH_3(CH_2)_{10}CO_2H$	acido dodecanoico	acido laurico
$CH_3(CH_2)_{12}CO_2H$	acido tetradecanoico	acido miristico
$CH_3(CH_2)_{14}CO_2H$	acido esadecanoico	acido palmitico
$CH_3(CH_2)_{16}CO_2H$	acido ottadecanoico	acido stearico

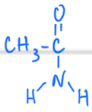
acido ottanoico e più grandi : ACIDI GRASSI



butandiammide
succinammide (*)

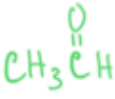
(*) Nome corrente accettato

ACETAMMIDE



H-CHO
formaldeide (*)
metanale

CH_3CHO
acetaldeide (*) più comune
etanale

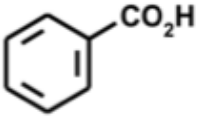


CH_3CO_2H ⇨ **CH_3CO_2K**

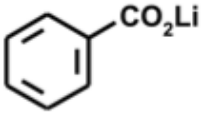
acido acet*ico* — acetato di potassio

$CH_3CH_2CH_2CO_2OH$ ⇨ **$CH_3CH_2CH_2CO_2Na$**

acido butano*ico* — butanoato *di* sodio



⇨



acido benzoico — benzoato di litio

cianuro di metile
acetonitrile
etanonitrile

$CH_3C{\equiv}N$;

COMPOSTI POLIFUNZIONALI

Se in una molecola sono presenti più gruppi funzionali, UNO SOLO viene scelto come gruppo funzionale ai fini del nome. Tutti gli altri vanno indicati come *sostituenti*, utilizzando i prefissi, in ordine alfabetico e, se necessario, i prefissi moltiplicativi.

La scelta del gruppo funzionale si effettua secondo il seguente ordine decrescente:

- **Sali di -onio**
- **Acidi carbossilici (acidi solfonici)**
- **Derivati degli acidi carbossilici: sali**
  - **esteri**
  - **alogenuri acilici**
  - **ammidi**
  - **nitrili**
- **Aldeidi**
- **Chetoni**
- **Alcooli, Fenoli (Tioli)**
- **Ammine**
- **Eteri (Solfuri)**

DOPPI LEGAMI

Gli alogeni ed il gruppo nitro sono *sempre* sostituenti.

TRIPLI LEGAMI
SOSTITUENTI